

Quick Start Guide

Get started with hyper2kvm in 5 minutes

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##

Prerequisites

- Linux system (Fedora, Ubuntu, RHEL, or SUSE) - Python 3.10 or later - Root/sudo access - Source VM disk files (VMDK, VHD, or raw images)

1. Install System Dependencies

Fedora / RHEL / CentOS Stream

```
sudo dnf install -y \  
python3 python3-pip \  
qemu-img qemu-kvm \  
openssh-clients rsync \  
libvirt-client libvirt-daemon-kvm
```

Ubuntu / Debian

```
sudo apt-get update  
sudo apt-get install -y \  
python3 python3-pip python3-venv \  
qemu-utils \  
openssh-client rsync \  
libvirt-clients libvirt-daemon-system
```

Verify Backend (Optional)

Note: VMCraft is the default backend and requires qemu-nbd and qemu-img.

```
# Verify qemu-nbd is available
qemu-nbd --version

# Verify qemu-img is available
qemu-img --version
```

2. Install hyper2kvm

Option A: Install from Source (Recommended for Development)

```
# Clone the repository
git clone https://github.com/ssahani/hyper2kvm.git
cd hyper2kvm

# Create virtual environment
python3 -m venv .venv
source .venv/bin/activate

# Install dependencies
pip install -U pip wheel setuptools
pip install -r requirements.txt
pip install -e .

# Verify installation
python -m hyper2kvm --help
```bash

Option B: Install from PyPI (When Available)

```bash
pip install hyper2kvm
hyper2kvm --help
```bash

3. Your First Migration

Scenario: Convert a Local VMDK to QCOW2

You have a VMware VMDK file and want to run it on KVM.

Step 1: Locate Your VMDK
```

```

'''bash
ls -lh /path/to/your-vm.vmdk
'''bash

Step 2: Run the Conversion

'''bash
sudo python -m hyper2kvm local \
 --vmdk /path/to/your-vm.vmdk \
 --flatten \
 --to-output /var/lib/libvirt/images/your-vm.qcow2 \
 --compress
'''bash

What this does:
- `local` - Process a local disk file
- `--vmdk` - Source VMDK path
- `--flatten` - Flatten snapshot chains
- `--to-output` - Output QCOW2 path
- `--compress` - Enable QCOW2 compression

Step 3: Verify the Output

'''bash
qemu-img info /var/lib/libvirt/images/your-vm.qcow2
'''bash

4. Common Scenarios

Linux VM with Network/Bootloader Fixes

'''bash
sudo python -m hyper2kvm local \
 --vmdk linux-vm.vmdk \
 --flatten \
 --to-output linux-vm.qcow2 \
 --fix-network \
 --fix-bootloader \
 --compress
'''bash

Windows VM with VirtIO Driver Injection

'''bash
First, download VirtIO drivers

```

```

wget https://fedorapeople.org/groups/virt/virtio-win/direct-downloads/latest-
virtio/virtio-win.iso

Convert with driver injection
sudo python -m hyper2kvm local \
 --vmdk windows-vm.vmdk \
 --flatten \
 --to-output windows-vm.qcow2 \
 --windows \
 --inject-virtio \
 --virtio-win-iso ./virtio-win.iso \
 --compress
```bash

### Fetch VMDK from ESXi and Convert

```bash
sudo python -m hyper2kvm fetch-and-fix \
 --host esxi.example.com \
 --user root \
 --remote /vmfs/volumes/datastore1/vm/vm.vmdk \
 --fetch-all \
 --flatten \
 --to-output vm.qcow2
```bash

### Fix Running VM Over SSH (No Conversion)

```bash
sudo python -m hyper2kvm live-fix \
 --host 192.168.1.100 \
 --user root \
 --sudo \
 --fix-network \
 --fix-bootloader
```bash

---

## 5. Using Configuration Files

Instead of long command lines, use config files:

### Create a Config File (vm-config.json)

```json
{
 "command": "local",
 "vmdk": "/data/vms/production-web.vmdk",

```

```

 "flatten": true,
 "to_output": "/var/lib/libvirt/images/production-web.qcow2",
 "compress": true,
 "fix_network": true,
 "fix_bootloader": true,
 "report": "/var/log/migration-report.md"
}
```bash

### Run with Config File

```bash
sudo python -m hyper2kvm --config vm-config.json
```bash

### Use Example Configs

```bash
Browse available examples
ls examples/json/

Use an example
sudo python -m hyper2kvm --config examples/json/10-local/local-linux-basic.json
```bash

---

## 6. Testing the Converted VM

### Test with QEMU (No LibVirt Required)

```bash
sudo python -m hyper2kvm local \
 --vmdk test.vmdk \
 --to-output test.qcow2 \
 --qemu-test \
 --dry-run
```bash

### Test with LibVirt

```bash
sudo python -m hyper2kvm local \
 --vmdk test.vmdk \
 --to-output test.qcow2 \
 --libvirt-test
```bash

```

```

### Manual Boot Test

```bash
Boot directly with QEMU
sudo qemu-system-x86_64 \
 -m 2048 \
 -smp 2 \
 -drive file=/var/lib/libvirt/images/your-vm.qcow2,if=virtio \
 -enable-kvm \
 -nographic
```bash

---

## 7. Deploy to Production

### Create LibVirt Domain

```bash
Generate domain XML
virt-install \
 --name your-vm \
 --memory 4096 \
 --vcpus 2 \
 --disk path=/var/lib/libvirt/images/your-vm.qcow2,format=qcow2 \
 --network network=default \
 --graphics vnc \
 --import \
 --print-xml > your-vm.xml

Define and start
sudo virsh define your-vm.xml
sudo virsh start your-vm

Check status
sudo virsh list --all
sudo virsh console your-vm
```bash

---

## 8. Common Issues and Solutions

### Issue: KVM permissions not configured

**Solution:**

```bash
Check KVM permissions

```



```

ls -l /dev/kvm
sudo usermod -aG kvm $(whoami)
Log out and back in

Load KVM modules
sudo modprobe kvm
sudo modprobe kvm_intel # or kvm_amd
```bash

### Issue: "Permission denied" errors

**Solution:**
```bash
Run with sudo
sudo python -m hyper2kvm ...

Or adjust permissions
sudo chown $(whoami) /var/lib/libvirt/images/
```bash

### Issue: VMDK not found

**Solution:**
```bash
Use absolute paths
sudo python -m hyper2kvm local \
 --vmdk "$(pwd)/vm.vmdk" \
 --to-output "$(pwd)/output.qcow2"
```bash

### Issue: Network doesn't work after migration

**Solution:**
```bash
Use network fixing
sudo python -m hyper2kvm local \
 --vmdk vm.vmdk \
 --to-output vm.qcow2 \
 --fix-network
```bash

### Issue: Windows won't boot

**Solution:**
```bash
Inject VirtIO drivers
sudo python -m hyper2kvm local \
 --vmdk windows.vmdk \

```

```

--to-output windows.qcow2 \
--windows \
--inject-virtio \
--virtio-win-iso /path/to/virtio-win.iso
```bash

---

## 9. Next Steps

### Explore Documentation

- [Installation Guide](01-Installation.md) - Detailed installation instructions
- [Examples](../examples/) - 30+ working configuration examples
- [Architecture](01-Architecture.md) - How hyper2kvm works internally
- [Troubleshooting](90-Failure-Modes.md) - Common problems and solutions

### Try Advanced Features

- Batch Migration - Convert multiple VMs
- vSphere Integration - Export directly from vCenter
- Cloud-Init - Prepare cloud images
- Custom Scripts - Post-processing hooks

### Get Help

- GitHub Issues: https://github.com/ssahani/hyper2kvm/issues
- Documentation: docs/` directory
- Examples: examples/` directory

---

## 10. Command Cheat Sheet

```bash
Basic conversion
sudo python -m hyper2kvm local --vmdk INPUT.vmdk --to-output OUTPUT.qcow2

With compression and flattening
sudo python -m hyper2kvm local --vmdk INPUT.vmdk --flatten --compress --to-output OUTPUT.qcow2

Linux with fixes
sudo python -m hyper2kvm local --vmdk INPUT.vmdk --fix-network --fix-bootloader --to-output OUTPUT.qcow2

Windows with VirtIO
sudo python -m hyper2kvm local --vmdk INPUT.vmdk --windows --inject-virtio --virtio-win-iso VIRTIO.iso --to-output OUTPUT.qcow2

```

```

Fetch from ESXi
sudo python -m hyper2kvm fetch-and-fix --host ESXI_HOST --remote VMDK_PATH --to-output OUTPUT.qcow2

Using config file
sudo python -m hyper2kvm --config CONFIG.json

Test conversion
sudo python -m hyper2kvm local --vmdk INPUT.vmdk --to-output OUTPUT.qcow2 --qemu-test

Dry run (preview)
sudo python -m hyper2kvm local --vmdk INPUT.vmdk --to-output OUTPUT.qcow2 --dry-run

Generate report
sudo python -m hyper2kvm local --vmdk INPUT.vmdk --to-output OUTPUT.qcow2 --report REPORT.md

Debug mode
sudo python -m hyper2kvm --log-level DEBUG local --vmdk INPUT.vmdk --to-output OUTPUT.qcow2
```bash

---

**You're ready to start migrating VMs! **

For detailed information, see the full documentation in the `docs/` directory.

### Advanced Examples

#### Example: Batch Migration

```bash
Create a list of VMs to migrate
cat > vms.txt <<VMLIST
/data/vm1.vmdk
/data/vm2.vmdk
/data/vm3.vmdk
VMLIST

Migrate all VMs
while read vmdk; do
 name=$(basename "$vmdk" .vmdk)
 sudo python -m hyper2kvm local \
 --vmdk "$vmdk" \
 --flatten \
 --to-output "/var/lib/libvirt/images/${name}.qcow2" \

```

```
--compress
done < vms.txt
```

### Example: Cloud-Init Injection

```
sudo python -m hyper2kvm local \
 --vmdk ubuntu-template.vmdk \
 --to-output cloud-ubuntu.qcow2 \
 --inject-cloud-init \
 --compress
```

## Troubleshooting

### Common Issues

#### Issue: Command fails with permission denied

**Symptoms:** - Error: "Permission denied" when accessing disk images - Cannot write to output directory

#### Solution:

```
Run with sudo
sudo python -m hyper2kvm --config your-config.yaml

Or fix permissions
sudo chown $(whoami) /path/to/output/directory
```

#### Issue: Guest disk mount fails

**Symptoms:** - Error: "guestfs\_mount: failed" or VMCraft mount error - Cannot inspect guest OS

#### Solution:

```
VMCraft is the default backend; check NBD availability
sudo modprobe nbd
ls /dev/nbd0

Check KVM permissions
sudo usermod -aG kvm $(whoami)
```

```
Log out and back in

Verify disk image
qemu-img info /path/to/disk.vmdk
```

For more issues, see [Failure Modes](#).

## Next Steps

Now that you've completed your first migration:

1. [Explore Examples](#) - 40+ ready-to-use configuration files
2. [Read the Cookbook](#) - Common migration recipes
3. [Understand Architecture](#) - How hyper2kvm works internally
4. [Windows Migrations](#) - If you need to migrate Windows VMs

## Getting Help

- **Issues:** [GitHub Issues](#)
- **Troubleshooting:** [Failure Modes Guide](#)
- **Documentation:** All docs in `docs/` directory